

Chapter 1

Introduction

One of the generation IV reactor designs is the fluid-fueled molten salt reactor (MSR) [91]. MSRs are advanced reactors intended to improve safety and efficiency over traditional reactor designs. Instead of a solid fuel, MSRs use a liquid fuel composed of a molten salt with some fissile nuclide, such as ^{235}U . MSRs have a wide variety of benefits: high operating temperature, low operating pressure, and high thermal conductivity and capacity of the salts. Other benefits include a high coefficient of thermal expansion for negative temperature coefficient of reactivity, high resource utilization through higher burn-up, reduced cost of fabricating and transporting fuel elements, and improved neutron economy by online fission product removal [167].

MSRs have several barriers to deployment which must be addressed before they are commercialized. Corrosion and material degradation is an issue, as the high temperature salts are highly corrosive. Another barrier to deployment is in reactor safety, security, and safeguards by design, as MSRs have distinct designs compared to traditional light water reactors. To properly design MSRs, models must be able to simulate them accurately. One specific phenomena of critical importance in MSRs are the circulating delayed neutron precursors (DNPs).

DNPs are nuclides which undergo β^- decay, producing a daughter, called an emitter, which emits a neutron [11]. Without properly modeling the DNPs in reactors, simulations may predict incorrect power responses, reactivity worth, and reactor controllability [44, 45, 96]. The DNPs are typically not modeled explicitly in simulations, but are instead represented using DNP group parameters, which are intended to have the same physical effect at a much lower computational cost. A brief historical timeline of DNP group parameter modeling advancements is given in Tables 1.1 and 1.2.

The progress in DNP modeling and group parameter formulation has progressed rapidly thanks to increased experimental data available for individual DNPs, which allowed for more accurate group parameter formulation. However, these formulations of DNP group parameters do not account for certain physics unique to MSRs. Specifically of interest is how the fuel movement and chemical reprocessing affects the formulation of DNP group parameters. Although there has also been much analysis on DNPs in MSRs, that work has not investigated this gap [200, 204, 100, 174, 95]. Most of these studies investigate movement of DNPs during transients. However, to have accurate transient models, it is necessary that the DNP group parameters are accurate. Thus, a new step forward for group parameter development is for DNP group parameters to accurately reflect the physical phenomena in MSRs.

The objectives of this research are to generate improved DNP group parameters that account for MSR physical phenomena, verify the parameters with the literature, validate the parameters by simulating transients and comparing with experimental data, and evaluate the impact of the improved parameters on safety, safeguards, and security by design. For example, the chemical removal of DNPs will lead to a decrease in the number of delayed neutrons

Table 1.1: Timeline of advancements in DNP modeling up to 1982.

1939	Roberts et al. discover delayed neutrons [147]; Bohr and Wheeler explain delayed neutrons [21].
1946	Burgy et al. conduct initial spectral measurements [31].
1947	Hughes et al. initial use of six DNP groups at Argonne heavy water pile [78].
1957	Keepin et al. generate improved DNP group parameters [89, 194].
1965	Amiel identifies seven DNPs that account for 80% of delayed neutrons [6].
1969	Marmol compiles experimental data on emission probability and half-life for 34 DNPs [46].
1972	Tomlinson maps 38 DNPs onto six groups [185].
1977	Saphier et al. construct delayed neutron spectra from 20 DNPs [162].
1982	Rudstam generate DNP group parameters and spectra using 67 DNPs based on half-lives [154].

emitted per fission event, on average. This will lead to a reduced total delayed neutron yield, which will in turn result in a decrease to the effective delayed neutron fraction, which corresponds to a shorter reactor period, thus reducing reactor controllability. The DNP group parameters currently in use for modeling MSRs may be overestimating the number of delayed neutrons in the system.

The physical phenomena in MSRs can be incorporated at varying levels of fidelity, with associated differences in cost and accuracy. In this document, three models are described: a 0D scaled model, which is already complete; a 0D flow model, which is proposed; and a spatially resolved model, which is discussed for future work. The least accurate and fastest is a 0D scaled model. This model neglects time dependence, decay chains, and cross-sections while scaling the flux and chemical removal rates based on the fraction of fuel salt in different regions of the primary loop. The intermediate approach is a 0D flow model. This model accounts for time dependence and decay chains, while applying the flux and chemical removal rates only in the relevant parts of the primary loop. Finally, the most accurate and slowest approach is a spatially resolved model. This model accounts for time and space dependence

Table 1.2: Timeline of advancements in DNP modeling from 1983 onward.

1983	England et al. generate DNP group parameters and spectra using 105 DNPs based on half-lives [54].
1988	Brady generates DNP group parameters and spectra using 271 DNPs using least squares [25]. Group parameters in ENDF/B-VI.8 set and remain unchanged through ENDF/B-VIII.0 [195, 29, 153, 23, 53, 24].
1999	Parish et al. state DNP group parameters in Evaluated Nuclear Data File (ENDF) are in need of revision [128].
2014	International Atomic Energy Agency (IAEA) Coordinated Research Project (CRP) on Development of a Reference Database for β -delayed neutron emission data [118].
2020	Leconte et al. use more than 400 DNPs to combine data with macroscopic experimental measurements [99].
2025	This study develops DNP group parameters for use in MSRs.

over the entire primary loop. The purpose of each model is to simulate irradiation of a fissile sample, from which DNP group parameters can be calculated. There are two types of irradiations considered: saturation irradiations, in which long-lived DNPs dominate, and pulse irradiations, in which short-lived DNPs dominate.

The preliminary results of this work cover impactful DNPs, sensitivity studies of the DNP group parameters, and verification of the 0D scaled model. The results show the total delayed neutron yield and average half-life values are within uncertainty bounds between the raw DNP data and the DNP group parameters. The match between the raw data and the group parameters shows the effect of each DNP is captured in the DNP group parameters. The sensitivity studies reveal each DNP can affect multiple DNP group parameters, not only the group with the most similar half-life to that DNP. The verification with the literature reveals that the 0D scaled model works well for longer-lived groups, but does not work well for shorter-lived groups. This is because the 0D scaled model only simulates saturation irradiations, which do not provide as much data for the shorter-lived DNPs as pulse irradiations. The other approximations do not lead to large differences with the literature, but the lack of decay chains results in

an inability for this model to treat pulse irradiations

The proposed work starts with development of the 0D flow model, which is followed by a verification stage to ensure the improved DNP group parameters are correct for a static case. This is then followed by validation of the model with a transient simulation of the Molten Salt Reactor Experiment (MSRE) with experimental data to ensure the improved DNP group parameters are physically accurate. The improved group spectra are then included in the formulation of the DNP group parameters, and verified for a static, or stationary, case. Finally, transients of the MSRE for various safety, security, and safeguards-related scenarios are simulated using the improved DNP group parameters and pre-existing group parameters to study their impact.

This set of proposed work is considered the critical path for a full analysis, though Section 5.4 presents non-critical work. The non-critical work includes development of the spatially resolved model, uncertainty tracking of MSR physical phenomena in the DNP group parameters, and investigation into the effects of varying the incident neutron energy spectrum on the DNP group parameters.